Measuring Firms' Exposure to Carbon Pricing: An Industry Concordance and Its Limits

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Research note, October 2026

Note. This note documents work I did as a research assistant on Professor Anh V. Persson's project on carbon pricing and corporate investment (Gies College of Business, May–August 2026). The concordance, the code, and every number reported here are my own work and can be reproduced from the accompanying repository. The project's empirical results are not reported. *Code and crosswalk tables: github.com/xiaokedai/carbon-pricing-concordance.*

1. The question

Studies of how carbon pricing affects firms usually identify the policy with an indicator: a firm is treated once its country has a carbon tax or an emissions trading system (ETS). The indicator is easy to build and easy to defend, but it treats a Swedish firm facing a tax above $100 per tonne and a Kazakh firm facing an ETS price of about $1 as the same event, and it treats a German steel mill inside the EU ETS and a German software firm outside it as the same firm. The alternative is to attach to each firm the price it actually faces. That requires two links that the indicator does not: the price has to be found for the firm's jurisdiction and year, and it has to be assigned to the firm's industry, because every pricing scheme covers some sectors and not others.

This note describes how I built the second link for a global Compustat-style panel, where the only industry information is a four-digit US SIC code, and what I found when I checked it. The main finding is simple to state. About a third of SIC codes do not map to a single priced industry. For most of those codes the ambiguity is harmless, because every candidate industry faces nearly the same price. For a specific minority, concentrated outside manufacturing, the candidates straddle the boundary of the pricing scheme, and the researcher's choice decides whether the firm is priced at all.

2. The price

Three cross-country price series were candidates: the World Bank Carbon Pricing Dashboard, which reports the headline rate of each instrument; the OECD net effective carbon rates, which add fuel excise taxes; and the emissions-weighted carbon price (ECP) of Dolphin and Merkle (2024), which weights each instrument's rate by the share of a jurisdiction's CO_2 emissions it covers. The three measure different things. In Canada in 2025, for example, the headline industrial benchmark rose to roughly $66 per tonne while the emissions-weighted price fell to about $10, because the consumer fuel charge was removed; both numbers are correct. I chose the ECP because coverage is the object of interest here: a rate that applies to no emissions is not a price the firm faces.

The ECP comes in two files. The national file gives one price per jurisdiction and year for 1990–2022, with separate values for taxes and ETSs, and it prices carbon at the subnational level where schemes are subnational: 11 US states, 13 Canadian provinces and territories, and 7 Chinese provinces. The industry file gives, for each of 164 GLORIA regions, each of 120 GLORIA industries, and each year from 1990 to 2021, the price that applies to that industry's emissions. Prices are in constant 2021 US dollars per tonne of CO_2. The industry file is the endpoint of the concordance below, and its one-year-shorter coverage means 2022 firm-years need the national price as a fallback.

I checked the national series cell by cell against the raw download for the alphabetical range I could pull in one file. Several hundred cells matched; the one exception was Estonia in 2025, where two pulls of the same series disagreed (0.53 versus 5.34), which I flagged rather than resolved.

3. Building the concordance

There is no published table from US SIC codes to carbon-pricing sectors. The defensible route is a chain of official concordances, each vintage-dated:

SIC 1987 → NAICS 2002 → NAICS 2007 → ISIC Rev. 4 → GLORIA industry → ECP price.

The first three steps use US Census Bureau concordance files (the 1987 SIC to 2002 NAICS table, the 2002 to 2007 NAICS revision bridge, and the 2007 NAICS to ISIC Rev. 4 table, the last cross-checked against the UN Statistics Division's correspondence tables). The fourth step is not a probabilistic concordance but an aggregation by construction: GLORIA's 120 industries are defined on ISIC Rev. 4, so each ISIC class belongs to exactly one GLORIA industry. The fifth step is a lookup in the ECP industry file by (GLORIA industry, region, year). An earlier version of the chain continued from GLORIA to IPCC emission-source categories; that step was dropped in July 2026 in favour of reading the industry price directly, which is simpler and is the level at which the data's authors report it.

I wrote the chain in Python and validated it end to end. All 1,005 SIC codes in the Census table (1,004 four-digit codes plus the Census row for auxiliary establishments) reach at least one ISIC class; all 420 ISIC classes reached map to a GLORIA industry; the 96 GLORIA industries reached match the sector names in the ECP file verbatim, so a price lookup cannot silently miss. Worked examples behave as they should: in Germany in 2019 an electric utility (SIC 4911) maps to $27.4 per tonne, a refiner (2911) to $25.8, a steel mill (3312) to $25.8, and a railroad (4011) to $0, because the EU ETS does not cover rail; a US utility maps to $0, because there is no national US price and the industry file does not carry the state schemes.

Where a SIC code reaches more than one GLORIA industry, the pipeline carries all of them. For a single assignment it uses a strict rule: the GLORIA industry reached by the largest number of ISIC classes for that SIC code, with ties broken by Census order. Section 5 shows what this rule gets wrong.

4. Where the mapping is one-to-many

The chain is one-to-many at two points. A SIC code can split into several NAICS and hence several ISIC classes: 552 of the 1,005 codes reach more than one ISIC class. Aggregating to GLORIA industries absorbs part of this, because different ISIC classes often belong to the same GLORIA industry, but 342 codes (34 percent) still reach more than one priced industry. Table 1 shows where they sit.

Table 1. SIC codes that map to more than one GLORIA industry, by SIC division

SIC division	SIC codes	One-to-many	Share
A Agriculture, forestry, fishing	58	16	28%
B Mining	31	13	42%
C Construction	26	13	50%
D Manufacturing	459	178	39%
E Transportation, communications, utilities	67	26	39%
F Wholesale trade	69	13	19%
G Retail trade	64	20	31%
H Finance, insurance, real estate	53	7	13%
I Services	150	52	35%
J Public administration	27	4	15%
All All divisions	1005	342	34%

Source: sic_naics_isic_gloria_master.csv; src/concordance_stats.py. One-to-many = the code reaches two or more GLORIA industries. The Census auxiliary-establishment row is omitted.

By count the problem is a manufacturing problem: 178 of the 342 codes are in Division D. By rate it is broadest in construction (50 percent) and mining (42 percent), followed by manufacturing and by transport, communications, and utilities (39 percent each), and narrowest in finance (13 percent) and wholesale trade (19 percent). Most ambiguous codes have two or three candidates (175 and 78 codes respectively); the tail is long, reaching 16 candidates for SIC 3999 (manufacturing industries not elsewhere classified) and 18 for SIC 7389 (miscellaneous business services). The codes at the tail are the residual categories whose titles already announce the problem.

Some of the ambiguity is a product of the concordance rather than of the firm. SIC 2097, manufactured ice, maps through NAICS 312113 to ISIC 3530, steam and air-conditioning supply, which GLORIA folds into electric power. SIC 7372, prepackaged software, reaches both "Publishing" and "Printing", and the strict rule picks Printing. Neither is wrong as a reading of the tables; both would surprise a researcher who looked only at the firm.

5. Does the choice matter?

A one-to-many code matters only if its candidate industries face different prices. I checked this directly. For each of the 342 codes and each of the 42 jurisdictions with a positive industry price in 2019, I collected the ECP of every candidate industry and compared the strict choice with the full set. Dropping candidates with no emissions in a region (the file records these as missing, not zero) leaves 13,698 code-by-jurisdiction pairs, 11,589 of them with at least one priced candidate. Table 2 summarises them.

Table 2. Dispersion of candidate prices for one-to-many SIC codes, 42 pricing jurisdictions, 2019

SIC division	Pairs	Candidates disagree on priced vs. not	Strict choice = 0, another candidate priced	Median range ($/t)	Median range ÷ max
A Agriculture	458	85%	49%	22.9	1.00
B Mining	459	6%	1%	1.8	0.09
C Construction	85	25%	0%	0.0	0.00
D Manufacturing	7372	6%	2%	1.2	0.07
E Transport/utilities	705	43%	22%	10.7	0.99
F Wholesale	384	46%	46%	11.7	0.92
G Retail	528	39%	15%	7.5	0.76
H Finance/real estate	112	0%	0%	0.0	0.00
I Services	1403	43%	33%	8.1	0.97
J Public admin	83	24%	1%	4.4	0.98
All	11589	19%	10%	1.7	0.12

Source: ECP industry file (Dolphin and Merkle 2024); src/price_dispersion.py. A pair is one SIC code in one jurisdiction with at least two candidate industries that carry a price row and at least one priced above zero. Range = highest minus lowest candidate price. Using $1 per tonne instead of zero as the threshold for "priced" raises the pooled disagreement share from 19 to 25 percent and the strict-zero share from 10.5 to 13.6 percent.

Two facts stand out. First, in manufacturing the choice almost never matters. Across 7,372 manufacturing pairs the candidates disagree on whether the firm is priced in 6 percent of cases, and the median gap between the highest and lowest candidate is $1.18 per tonne, about 7 percent of the maximum. The German steel code is typical: its four candidates (basic iron and steel, coke, refined petroleum, sawmill products) are priced between $14 and $26, all inside the ETS. Second, outside manufacturing the choice matters a great deal. In transport and utilities, wholesale, retail, and services, the candidates disagree on whether the firm is priced in 39 to 46 percent of pairs, and the median gap is $7 to $12 per tonne, which is most of the maximum. These are the codes whose candidates straddle the coverage boundary of an ETS: electricity generation is inside, water supply is outside; printing is inside, publishing is outside; road freight is outside the EU ETS but inside Sweden's and France's carbon taxes, at $124 and $50 per tonne in 2019. For these codes the strict rule assigns a zero price while another candidate is priced in 15 to 46 percent of pairs.

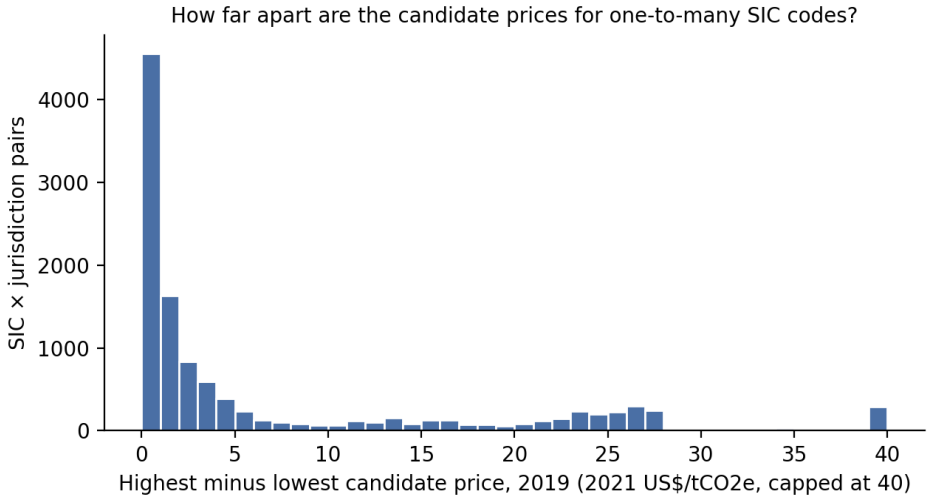


Figure 1. Gap between the highest and lowest candidate price for one-to-many SIC codes, 2019, pooled over the 42 pricing jurisdictions (11,589 pairs with at least one priced candidate; gaps above $40 shown at $40). The mass at zero is manufacturing; the right tail is the coverage boundary.

The pattern also differs by instrument. In jurisdictions whose price comes only from an ETS (most EU member states, the United Kingdom, Japan, Kazakhstan, Chile, Colombia), 74 to 78 percent of pairs have any priced candidate and the candidates disagree on coverage in 22 to 24 percent of them. In jurisdictions with an economy-wide carbon tax alongside the ETS (Sweden, Finland, France, Norway, Denmark, Ireland, Portugal, Switzerland), almost every candidate is priced (99 percent) and disagreement on coverage is rare (4 to 7 percent), but the median gap between candidates is large where the tax is high ($8 to $18 per tonne in the Nordic countries and France), because the tax rate differs by fuel and sector. An ETS makes the one-to-many choice a question of whether; a carbon tax makes it a question of how much.

Table 3 shows the strict rule's failures in one jurisdiction. For SIC 1311, crude petroleum and natural gas, the modal candidate is "Hard coal", priced at $6.6 in Germany in 2019, while the two candidates a reader would expect, gas and petroleum extraction, are priced at $27 and $25. For SIC 7372 the rule gives software firms the price of a print shop. In both cases each candidate is reached by exactly one ISIC class, so the modal rule is a tie broken by the order of the Census table; the rule reads positions, not titles. Both are fixable by hand for the codes that matter, and the repository carries an override file for them; the point of the table is that a mechanical rule needs that audit.

Table 3. Worked examples, Germany 2019

SIC code	Candidate GLORIA industries (* = strict choice)	ECP, Germany 2019 ($/t)
3312 Steel works, blast furnaces	Basic iron and steel*; Coke oven products; Refined petroleum products; Sawmill products	25.8*; 14.1; 25.8; 24.9
1311 Crude petroleum and natural gas	Hard coal*; Gas extraction; Petroleum extraction	6.6*; 26.9; 24.8
2819 Industrial inorganic chemicals, n.e.c.	Basic organic chemicals*; non-ferrous metals; dyes and paints; gas extraction; fertilizers; other non-metallic minerals; administrative services	24.5*; 25.6; 24.8; 26.9; 23.5; 24.4; 0.0
3711 Motor vehicles and car bodies	Motor vehicles*; Machinery and equipment; Other transport equipment	24.2*; 24.1; 25.0
7372 Prepackaged software	Printing*; Publishing	26.2*; 0.0
4213 Trucking, except local	Road transport (single candidate)	0.0
2097 Manufactured ice	Electric power generation (single candidate)	27.4

Source: src/price_dispersion.py, table6_examples_2019.csv. Prices in constant 2021 US$ per tonne of CO_2.

6. What this means for measurement

The exposure variable a researcher would carry into a regression is a price assigned to a firm through its SIC code. For two thirds of codes the assignment is unique, and for most of the remaining third the candidates agree to within a few dollars, so the error from the choice is small and roughly classical. For the codes at the coverage boundary the error is neither small nor classical: the strict rule sets the price to zero for a firm that may be fully covered, and it does so for whole industries (utilities, trade, services) rather than at random. Measurement error of that kind is correlated with sector, and sector is correlated with the investment and financing behaviour that studies of carbon pricing are about. Whether it attenuates an estimated response or biases it in a particular direction depends on the specification, and I have not tested that; what the concordance makes possible is to test it, by estimating the same specification under the strict mapping, under a broad mapping that averages or spans the candidates, and

on the sample that drops the ambiguous codes altogether. The 342 codes and their candidates are tabulated in the repository for that purpose.

Three further limits of the measure are independent of the one-to-many problem. A principal SIC code describes a firm's main activity, not its activity portfolio, so a diversified firm is priced as if it were its largest segment. The ECP weights by CO_2 only; mining codes, whose emissions are largely methane, therefore look nearly unpriced even where methane is taxed. And the industry file does not carry the subnational schemes that the national file does, so a Californian utility is priced at zero at the industry level and at the state price at the national level; which to use is a choice about the research question, not about the data.

7. What I did not do

I did not merge the concordance to firm data or estimate anything; the firm-level merge is the project's next step and belongs to its authors. The numbers in Sections 4 and 5 are properties of the concordance and of the public ECP file, computed with the scripts in the repository, and can be rerun for any year from 1990 to 2021 and any threshold for "priced". The strict rule is a reproducible default, not a recommendation; for the codes in Table 3 and their peers a title-based override is better, and the repository flags them for that purpose.

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